

Performance Testing Report - LaunchPad AI

1. Objective

The objective of performance testing is to verify that LaunchPad AI performs efficiently under normal usage conditions. The testing focused on responsiveness, API communication, usability, and overall application performance.

2. Testing Environment

Operating System: Windows 11
Backend: Flask
Frontend: HTML, CSS, JavaScript
AI Service: Google Gemini API
Browser: Google Chrome

3. Test Scenarios

The application was tested with multiple startup ideas from Healthcare, Education, Agriculture, FinTech, and E-commerce domains. Invalid inputs, long descriptions, and repeated requests were also evaluated.

4. Test Results

Test Case	Expected Result	Status
Load Home Page	Loads successfully	Passed
Submit Startup Idea	Data accepted	Passed
Gemini API Response	Analysis generated	Passed
SWOT Analysis	Displayed correctly	Passed
Market Analysis	Displayed correctly	Passed
Growth Strategy	Generated successfully	Passed
Invalid Input	Validation message shown	Passed
Responsive Design	Works on different screen sizes	Passed

5. Performance Summary

The application responded smoothly during testing. AI-generated reports were produced successfully after receiving responses from the Gemini API. User interactions remained responsive and the interface functioned correctly across tested scenarios.

6. Conclusion

Performance testing confirmed that LaunchPad AI meets its functional objectives. The platform successfully validates startup ideas, generates business insights, and provides a responsive user experience. Future improvements may include caching, database integration, and deployment optimization.